Robert Bamler



AI Scientist & Research Engineer • PhD in Theoretical Physics

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I'm a machine-learning scientist and engineer with 10+ years of experience, five of them as a **professor of machine learning** (>1,900 citations). I specialize in **probabilistic ML & uncertainty quantification**, **AI for science**, and **efficient AI** (quantization, compression, inference acceleration). I also bring **industry R&D experience** (Disney, Nokia; patents) and a PhD in theoretical physics, and remain a **hands-on developer** (Python/PyTorch, Rust, C++, GPU compute shaders).

My technical edge: I derive machine learning methods from a deep understanding of the mathematical models, and I implement them to run fast on real hardware.

Experience

Sabbatical

06/2026 – present Munich, Germany

- **Focus:** choosing my next role carefully; learning agentic software engineering hands-on by building the open-source library kladde (Rust/C++).

Principal Researcher

Nokia TECH, Visual AI Systems Research

11/2025 – 05/2026 Munich, Germany

- **Context:** Nokia announced the closure of its Munich site shortly after I joined; I left on good terms (see references).
- **Inventions:** lead inventor of 2 and co-inventor of 1 more method for efficient AI model compression; built the demos myself and initiated patent filings and standardization in the MPEG-AI consortium.

Professor of Machine Learning and Data Science

University of Tübingen, Cluster of Excellence "Machine Learning: New Perspectives for Science" & Tübingen AI Center

11/2020 – 10/2025 Tübingen, Germany

- **Leadership & funding:** founded & led a group of 9 researchers (5 PhD, 2 MSc, 2 BSc); won EUR 1.2M in highly competitive funding (DFG Emmy Noether grant) for research on resource-efficient ML & AI for science.
- **Research:** AI for science (hybrid physics/ML models for thermodynamics), uncertainty quantification & OOD detection, generative models, and model compression (NeurIPS, ICML, ICLR, *Chem. Sci.*).
- **Hands-on:** wrote and maintain the library constriction; upstreamed GPU features to wgpu; prototyped GPU kernels for faster LLM inference.
- **Teaching:** created a new course on AI-based data compression (videos).

Postdoctoral Research Associate

University of California, Irvine

12/2018 – 10/2020 Irvine, CA

- **Research:** faster training for large output spaces (ICLR), model compression & quantization (ICML), and ML for thermodynamics that beat the established physical model (*J. Phys. Chem. Lett.*).
- **Mentoring:** co-advised 2 PhD students and 1 intern.

Postdoctoral Research Associate

Disney Research & Development, Walt Disney Imagineering

09/2016 – 12/2018 Pittsburgh, PA ('16–'17) & Los Angeles, CA ('18)

- **Applied AI/ML:** ML systems for theme parks and TV shows (granted US patent); published at NeurIPS & ICML.
- **Cross-functional AI strategy:** advised other business units on AI/ML.

Skills

Expertise:
AI for science • hybrid physics/ML models • uncertainty quantification • OOD detection • generative models • statistical physics • model compression & quantization • inference acceleration

Programming:
Python • PyTorch • TensorFlow • HF transformers • Rust • C++ • CUDA • GPU shaders (WGSL, MSL, SPIR-V, HLSL)

Engineering practice:
Linux • Docker • Git • CI (GitHub Actions) • testing & benchmarking • AI coding agents

Leadership & strategy:
mentoring • team coordination • teaching • advising leadership on emerging opportunities

Goals

Research and engineering in a team of experts, at the intersection of machine learning, physics, and efficient computation (e.g., foundation models for physical simulation or fast LLM inference).

Education

- **PhD in Theoretical Physics** 07/2016
  University of Cologne, Germany
  (statistical many-body theory; advisor: Prof. Achim Rosch; supported by Deutsche Telekom Foundation)
- **Diploma in Physics** 12/2011
  Technical University of Munich, Germany
  (equivalent to an MSc; incl. exchange year at Universidad de Granada, Spain)

References

- **Prof. Stephan Mandt**, Associate Professor, University of California, Irvine;
  Relationship: my supervisor at Disney Research and UC Irvine (09/2016–10/2020).
- **Hamed Rezazadegan Tavakoli**, Head of Visual AI Systems Research at Nokia;
  Relationship: my manager at Nokia.

Selected Open-Source Contributions

- **wgpu** (5 merged pull requests, 2025; `github.com/gfx-rs/wgpu`): implemented WGSL's packed 4×int8 dot products in wgpu's shader compiler for all backends, using native GPU instructions on Vulkan and DirectX 12 and vectorized code on Metal—a building block for int8-quantized inference on consumer GPUs (wgpu is the library that powers WebGPU in Firefox).
- **constriction** (own project, 2020–present; `bamler-lab.github.io/constriction`): Rust/Python library of fast entropy coders for AI-based compression; 100+ GitHub stars; tested, benchmarked, and released via CI.
- **GPU kernels for compressed neural networks** (proof of concept, 2024–25; `github.com/bamler-lab/compressed-nn-ops-demo`): GPU kernels for multiplication with entropy-coded weights, decoding on the fly to cut GPU memory traffic in LLM decoding.
- **kladde** (own project, 2026–present; `kladde-dev.github.io`): durable data structures (think "SQLite without SQL"); spec-driven development with reference implementations in Rust and C++, and a demo integration into Inkscape (≈ 5 µs overhead per edit).

Patents

- S. Mandt and **R. Bamler** (assigned to Disney Enterprises, Inc.), *Dynamic word embeddings*, US11068658B2 (granted)
- *(further patent applications from my time at Nokia are pending)*

Selected Publications

For a full list, please see Google Scholar; most papers contain a link to a GitHub repository with the code.

AI for Science: Hybrid Physics/ML Models

- J. Zenn, D. Gond, F. Jirasek, and **R. Bamler**, *Balancing Molecular Information and Empirical Data in the Prediction of Physico-Chemical Properties*, Digital Discovery 4 (2025)
- F. Jirasek, **R. Bamler**, and S. Mandt, *Hybridizing Physical and Data-Driven Prediction Methods for Physicochemical Properties*, Chem. Commun. 56 (2020)
- F. Jirasek, R. A. S. Alves, J. Damay, R. A. Vandermeulen, **R. Bamler**, et al., *Machine Learning in Thermodynamics: Prediction of Activity Coefficients by Matrix Completion*, J. Phys. Chem. Lett. 11 (2020)
- T. Cinquin, S. Lo, F. Strieth-Kalthoff, A. Aspuru-Guzik, G. Pleiss, **R. Bamler**, et al., *What Actually Matters for Materials Discovery: Pitfalls and Recommendations in Bayesian Optimization*, ICLR 2025 Workshop on AI for Accelerated Materials Design

Uncertainty Quantification & Out-of-Distribution Detection

- T. Cinquin, M. Pförtner, V. Fortuin, P. Hennig, and **R. Bamler**, *FSP-Laplace: Function-Space Priors for the Laplace Approximation in Bayesian Deep Learning*, NeurIPS 2024
- T. Cinquin and **R. Bamler**, *Well-Defined Function-Space Variational Inference in Bayesian Neural Networks via Regularized KL-Divergence*, UAI 2025
- T. Z. Xiao, J. Zenn, and **R. Bamler**, *The SVHN Dataset Is Deceptive for Probabilistic Generative Models Due to a Distribution Mismatch*, NeurIPS 2023 Workshop on Distribution Shifts

Efficient AI: Compression, Quantization & Faster Training

- A. Conzelmann and **R. Bamler**, *Reducing Storage of Pretrained Neural Networks by Rate-Constrained Quantization and Entropy Coding*, arXiv:2505.18758 (2025)
- Y. Yang, **R. Bamler**, and S. Mandt, *Variational Bayesian Quantization*, ICML 2020
- **R. Bamler** and S. Mandt, *Extreme Classification via Adversarial Softmax Approximation*, ICLR 2020

Physics-Inspired Probabilistic Machine Learning

- J. Zenn and **R. Bamler**, *Differentiable Annealed Importance Sampling Minimizes The Jensen-Shannon Divergence Between Initial and Target Distribution*, ICML 2024
- **R. Bamler**, C. Zhang, M. Opper, and S. Mandt, *Tightening Bounds for Variational Inference by Revisiting Perturbation Theory*, J. Stat. Mech. (2019)
- **R. Bamler** and S. Mandt, *Improving Optimization for Models With Continuous Symmetry Breaking*, ICML 2018
- **R. Bamler**, C. Zhang, M. Opper, and S. Mandt, *Perturbative Black Box Variational Inference*, NeurIPS 2017

Theoretical Physics (PhD and Diploma Research)

- **R. Bamler** and A. Rosch, *Equilibration and Approximate Conservation Laws: Dipole Oscillations and Perfect Drag of Ultracold Atoms in a Harmonic Trap*, Phys. Rev. A 91 (2015)
- F. Freimuth, **R. Bamler**, Y. Mokrousov, and A. Rosch, *Phase-Space Berry Phases in Chiral Magnets: Dzyaloshinskii-Moriya Interaction and the Charge of Skyrmions*, Phys. Rev. B 88 (2013)
- A. B. Sichert, **R. Bamler**, and J. L. van Hemmen, *Hydrodynamic Object Recognition: When Multipoles Count*, Phys. Rev. Lett. 102 (2009)